

Visualizing the Cost Function $J(w, b)$

To understand the cost function for the full linear regression model (with both w and b), we need to visualize it in three dimensions.

The 3D Surface Plot

- When we have two parameters, w and b , the cost function $J(w, b)$ can be plotted as a **3D surface**.
 - The two horizontal axes represent the parameters w and b .
 - The vertical axis (the "height" of the surface) represents the cost $J(w, b)$.
 - The shape of this surface for linear regression is a **bowl shape** (a convex surface).
 - Our goal is to find the single point at the **very bottom of this bowl**, as this point corresponds to the values of w and b that result in the minimum possible cost.
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The Contour Plot

A **contour plot** is a convenient way to visualize a 3D surface in a 2D graph. It's like looking down at the 3D bowl from directly above.

- **Analogy:** Think of a topographical map of a mountain, where each line on the map represents a specific elevation.
- **How it works:**
 - The two axes are the parameters w and b .
 - Each oval (or ellipse) on the plot connects all the points (w, b pairs) that have the **exact same cost** $J(w, b)$.
 - These ovals are like horizontal "slices" of the 3D bowl.
- **Finding the Minimum:** The minimum of the cost function is located at the **center of the innermost oval**.